

Comparative effects of tic-tac-toe exercise versus squat jumps on speed, agility, and explosive power among fast bowlers (RCT).

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DEDICATION

Firstly, this research project is dedicated to Allah Almighty and His Rasool (S.A.W).

Secondly, to our parents, whose unwavering love and support have been a driving force behind our academic pursuits. We are forever grateful for their belief in us and their encouragement, which worked as a constant source of inspiration for us in order to accomplish this attempt.

We also dedicate this project to our beloved teachers and ourselves.

DECLARATION

The effort described in this research was fulfilled by us under the supervision of Dr. HALIMA SHOUKAT from the Department of the COLLEGE OF PHYSICAL THERAPY by GOVERNMENT COLLEGE University, Faisalabad, Pakistan. We proclaim that the subject of the thesis is **COMPARATIVE EFFECTS OF TIC-TAC-TOE VERSUS SQUAT JUMPS ON SPEED, AGILITY, AND EXPLOSIVE POWER AMONG FAST BOWLERS**. The parts of the thesis are the result of our research, and no part has been copied from any printed source (excluding the citations, standard mathematical or genetic models/ equations/ formulas/ agreements, etc.). We proclaim that this work has not been presented for any degree/ diploma award. The university may take steps if the details given are incorrect at any phase.

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We hope that our innovative work will help to contribute to the ever-advancing knowledge of research.

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Abstract:

Introduction: Cricket fast bowlers deliver the ball consistently at high speeds over long bowling spells, which requires highly coordinated and well-planned movement solutions. Speed, agility, and explosive power are the capabilities required by fast bowlers to show rapidity and promptness in whole body movement with change in velocity and direction against the stimulus. The integration of specific drills, such as tic-tac-toe drills and squat jumps, into the training regimens of fast bowlers can be used to enhance speed, agility, and explosive power. Tic-tac-toe drills, which emphasize promptness and rapid footwork while squat jumps are one of the plyometric exercises, focus on the lower body strength and power, and can be effective in a fast bowler's performance enhancement. **Aims and Objectives:** This study examined the effect of tic-tac-toe exercises and squat jumps on speed, agility, and explosive power among fast bowlers. **Material and Methodology:** A total of 42 participants were selected between the ages of 18-30 years based on inclusion criteria from different clubs and cricket academies of Faisalabad. The purposive sampling technique was used to collect the sample. By coin tossing, participants were divided into two groups, A and B. Group A performed the tic-tac-toe exercise, while Group B performed squat jumps on alternate days for 6 weeks. Both groups followed full-body dynamic stretching along with training regimens as a baseline therapy. Speed, agility, and explosive power were tested before and after training sessions by using the 30-metersprint test, T-test, and vertical jumps, respectively. The results of the study showed that for improvement in the 30-meter sprint test and vertical jump test, the squat jumps regimen was superior to tic-tac-toe exercise, while for the agility T-test, the squat jumps regimen and tic-tac-toe exercise showed equal effects.

Keywords: Agility training, Explosive power, Fast bowlers, Speed, Squat jumps, Tic-tac-toe

CHAPTER # 1

INTRODUCTION

Introduction:

Playing sports has a big effect on youth as it strengthens their morals, builds their discipline, teaches them sporting values, and improves their achievements, leading to national pride. There is an impact and influence of the sports sector to improve economic development on a micro and macro basis(Pranata et al., 2023).Playing sports is not only a physical activity but also plays an important role and a part of socialization and adaptation to society. Sports are a part of culture and the best performances can be made only through the planned, systematic and controlled training systems. (Dilnoza, 2023)

One of the striking features of twentieth-century life is sport, as evidenced by the variety and popularity of sporting events in the most diverse parts of the world. Sports roles cover a large number of professions and responsibilities, from the direct involvement in athletic competitions to support the broader sports industry. This new phenomenon in society has attracted the interest of researchers in many fields of sports involving sports training. Sports are an important contributor to the human pacified nature. (Anandakumar et al.)

Sports training is very important and must be included in every athlete's daily schedule. Training over time helps the body get stronger and more enduring, makes skills better and increases motivation, ambition, and confidence. Being involved in sports training helps athletes become more involved in their sport activities and understand the significance of being both physically and mentally healthy. Sports training is also useful for athletes because it helps them think positively, which improves their focus and self-esteem. (Uppal, 2021)

Among all sports, cricket is an internationally acknowledged bat-and-ball game played for eleven-a-side teams on a 22-yard pitch. One group begins by batting, trying to score runs by hitting the ball, and the other team throws and fields. The team bowling aims to keep the batting team from getting as many runs as they can. Fast bowlers and slow bowlers make up the bowling team, which helps keep runs off the board for the batting team. A fast bowler tries to bowl as fast as possible to make it hard for the batsmen to see and react to the ball.(Ramachandran et al., 2024).

Sport known for periods of strength and speed separated by rest is cricket. Many important situations in cricket involve players moving quickly or explosively. Some examples are fast bowling, jumping to take a good catch, running to score or prevent a boundary or hurling the ball at the bowling crease to get someone out. Studies have found these components important, so requirements are put in place for both lower-body and upper-body power. It is obvious that strengthening and power for the lower body should be part of the training planning.(Herridge et al., 2020)

Cricket players are usually required to have great athleticism, strong arm power and endurance to run and bowl for long periods. Cricket is different since matches are played in 3 formats: First-class, List-A, and Twenty20 (T20). They each use different parts of the body physically. Players are grouped as batters and bowlers, and the bowlers may be either seam or spin. In ONE-DAY matches, each innings goes up to 50 overs, and bowlers get to bowl a maximum of 10 overs. So, since fast bowlers bowl a lot each season, they are more likely to get injured.(Webster et al., 2022)

Fitness in cricket has not always been as important as it is in many other sports. There is no exaggerating how important fitness is in all activities. A learning person performs better in sports if they are physically active. Besides being tactical, cricket also requires athletes to be in good physical shape, mentally strong, and have effective gamesmanship. All aspects of cricket depend on motor fitness, especially attacking fielding, fast bowling, overarm throws, running, and hitting..(Boby, 2023a)

Experts think sports such as cricket require players to use their bodies to complete physical activities involving motor control, jumping, bending and stretching. Since cricket has a lot of action, the team and each player have to respond to changes quickly. Being strong, powerful, and fit impacts many cricket roles such as hitting the ball, catching it, and running soon after. Strong motor abilities can result in performing faster while running between wickets, more powerful hits, and more accurate throws. Sports training decreases the possibility of injury. For maintaining motor fitness, players should train with exercises that improve strength, endurance, flexibility, and power. (Boby, 2023b)

Athletes need to be fast, agile and very powerful in cricket, as they must constantly adjust to various speeds, jump, sprint, catch, dive and make quick movements that don't take too much time (such as batting, bowling, fielding, wicket keeping, making a run). More agility and explosive power usually lead to a better athletic performance. Strong legs and rushing power (speed and fast steps) are essential for cricketers. They each have a role in affecting the speed of execution. It matters as well for bowlers. They require agility and strength as well to help them keep track of and strengthen their capacity to accept the stresses put on the legs by each step delivery.(Ghosalkar et al., 2020)

In a professional cricket match, players execute several explosive bursts, including running, tackling, jumping, shooting, and changing tempo. Therefore, a person must use alertness and movement speed to do these tasks in cricket. Some workout methods help agility by changing the way neuromuscular adaptation, muscle spindles, Golgi-tendon organs, tendons, joints, balance, and body position function. For talented athletes, fast, strong muscles allow them to jump as high as possible. Doing plyometric exercises helps athletes become faster, more agile, and more explosive. Neuromuscular training or plyometric training, quickly improves how much power you can generate in your muscles. (Weldon et al., 2021)

If the bowler is accurate, batters may struggle to score for a long period which may rise the pressure to score and lower the accuracy of stroke-making from the batter. Pace bowlers should run 15-30 meters, jump into the air, land with one foot behind and one in front and do it all with a wide-bit run-up like baseball pitchers and javelin throwers. Speed in the winds-up and an accelerated speed towards the bowling crease do generally contribute to throwing the ball at a greater speed. (Feros et al., 2020)

The intermittent burst of speed and power is a part of cricket. Looking at game-changing scenarios in cricket, which are in the hands of fast bowlers, requires speed and explosive power. Well-known game-changing situations are fast bowling and taking a dive to catch the ball. Intermittent sprinting happens during fast bowling (at least 3 sprints with only 50 to 60 seconds break between them) during matches which is why anaerobic power is essential. With the particular type of strength and power training, there is a significant improvement in sprinting capabilities.(Herridge et al., 2020)

While players may require a variety of physical fitness attributes for the game-play demands, an important element required to successfully play a game includes sports-specific training. Because sports-specific fitness matters, athletes must be fit for the particular sport they join. For example, many sports require muscle strength and explosive power for better athletic performance. Through special physical training, the required skills for the sports can be improved. How extended, strong, fast and agile athletes are will determine their achievements.(Farley et al., 2020)

Fast bowlers (also known as pace bowlers) have to switch directions frequently during the game and balancing the need for speed in bowling. They use their bowling speed, accuracy, and consistency for dismissals. Having a fast ball speed helps the bowler, since it allows less time for the batsman to decide how to hit the ball. A ball being delivered at speed 44.4 m/s will take the batsman roughly 0.44 seconds to catch. If the bowlers continue bowling quick consistently, the batsmen do not settle, and if the pace drops, it helps the bowler. Speed and agility are the critical physical attributes that contribute to the fast bowler's performance. (Boyat & Rathod, 2021)

How fast the bowler runs up affects how fast the ball will come off the hand. Increasing the run-up creates more momentum which the bowler passes through their body into the bowling action, quickly making the ball move. Leg movement and transferring energy up through their hips, shoulders and arm allow bowlers to deliver quicker pitches. The return path should be stable to avoid disturbing your balance and tempo, which makes you go faster and without accuracy. A powerful and well-planned running approach before bowling gives the bowler the chance to send the ball with great velocity and control. For this reason, player's run-up velocity is important for generating speed. (Ghosh & Mondal, 2024)

Attaining maximum ball release speed is the most important goal for fast bowlers, as previously discussed, because it gives batsmen less time to evaluate the trajectory of coming short. Fast bowlers should make an explosive leap before finishing the run-up in the delivery stride to develop optimal ball release speed (BRS). Back foot contact (BFC) and front foot contact (FFC), each with ground reaction forces, happen during the final part of a player's delivery stride. Many researches explain that the high forces bowlers apply

on the ground when their front foot lands are key for their bowling performance. So, lower limbs unilateral or bilateral strength training is required to increase BRS for fast bowlers. (Callaghan et al., 2024)

Fast bowlers have to take a faster run-up, and for a faster run-up, they require agility too, particularly in terms of generating power and maintaining balance. Agility is one of the critical components of any sport. In cricket, fast bowling is an action where agility is an important component to excel. It is linked with teachable physical qualities like strength, power, and balance. It includes the aggregate capacities of a player, which helps the athlete to perform a group of movements with better quality and effect. These involve their ability to react, balance, orient themselves, and to perceive a rhythm of movement to perform a particular task precisely. (Jayawant & Kuttan, 2022)

Being able to move quickly, stay consistent on your feet and change direction without losing balance and control is called agility. Having agility involves balance, coordination, strength and speed. Having good agility helps professional, recreational and tactical athletes switch directions during sports. As your agility increases, you become more stable when you move fast, you gain control over your muscles and you are less likely to be injured. (Bin Shamshuddin et al., 2020)

Being able to change direction quickly is an important element of agility and plays a big role in many sports and activities. During a sports competition, changing direction quickly matters a lot since it gives an edge to those who do it fastest. The rate at which a sprinter can change direction depends on how fast they run and the muscle development in their legs which is shaped by their training. One must quickly use force to make speed easier to gain and to control speed when changing direction. (Suarez-Arrones et al., 2020)

The players in the field have to do specific and quick movements during the game. So, players should have strong physiological and neuromuscular coordination, as well as a high level of lower limb strength to prevent injuries. To improve physical fitness and prevent injuries, a recent and popular form of exercise, which is time-saving and very efficient, is used. This new technique is called as plyometric training, which enhances strength and explosive power, which can help athletes to move quickly. These quick movements are

usually associated with agility. So, plyometric training for 7-8 weeks can improve agility.(Latif et al., 2023)

Changes in direction by players in team sports tend to go along with greater strength and speed. Change of direction speed improvements usually go hand in hand with vertical and countermovement jumps. In addition, comparing how change of direction speed is affected by the dominant and non-dominant legs showed that strong leg muscles support the ability to turn quickly. It therefore seems true that doing these exercises enables a player to display greater strength in turning, resulting in a lower center of mass, a stronger stop, a shorter time in contact with the ground and higher horizontal and vertical dynamic forces in the acceleration step. (Keller et al., 2020).

Cricket depends on agility which combines speed, coordination and balance. Better Agility makes it easier for your body and brain to communicate which also lessens the chance of injury. Moving fast between wickets and precisely catching and passing the ball are skills you need. Being agile, flexible and strong is vital for any athlete in cricket (batsmen, spin bowlers, fast bowlers). (Jayawant & Kuttan, 2022)

How quickly an athlete can go is influenced by their starting speed, rate of acceleration and highest speed. There has been a noticeable growth in interest in functional training since it helps with athletic performance. This way of training improves the target movement, many times by reproducing the movement instead of dedicating the exercise to a single muscle. It supports better sprinting and jumping abilities. Many studies have found that most traditional training programs miss out on multiarticular and multiplanar training which is important for performance in sports. Training for a specific purpose encourages better results in those same areas. (Bashir et al., 2022)

Gamified learning has become more popular nowadays, along with the specific training and functional training methods. Some gamified models involve modern technology while others do not. Tic-Tac-toe exercise is one of the gamified exercises involving strategy and quick decision making. A tic-tac-toe game requires athletes to move through different marks, put cones in the given areas, shoot some lay-ups and get back to the starting spot. It needs individuals to be skillful, quick and accurate through movements that are both

tactical and energetic. The use of tic-tac-toe exercise in training regimens has shown the improvement in speed, agility and accuracy in athletic performance. (Sitanggang, 2024)

Traditionally, playing tic-tac-toe is done on a board shaped as a three-by-three grid. In this game, two players X and O place their mark in the empty area one after another. If someone can line up three of their matching pieces in a row (up, down or diagonally), they win the game. The game ends when each of the nine squares is full. Players must keep going until a player lands three icons in a row which decides the game. The move of tic-tac-toe encourages a player to use the idea of taking turns in their attention. Thus, the game changes the focus from aiming to win to aiming to not lose. (Shakow)

Physically, game playing requires marking a tic-tac-toe grid on the ground and using six cones, three cones in one color and three in the other. The teams stand in a line far from the grid and after the coach calls “GO,” both players run across towards each other, facing away from the grid. Now, each player picks an area to place the cone, hoping to make it hard for the other team to get to three in a row (like you do in regular tic-tac-toe). After their action, they move back to the beginning as fast as they can. Players compete until the first finishes the game. ("Tic-Tac-Toe,")

Before training for sports, the models focus on helping young athletes improve basic movement abilities. Rather, in many sports, these two kinds of skills are combined. Certain sports, like parkour, can work on both aspects by boosting movement skills and athletic performance. Parkour tic-tac showed higher acceleration and less power from the lower limb when compared to other techniques. This approach can lead to better athletic coaching that joins motor learning, performance and development of skills. (Mark D Williams et al., 2024)

A good physical condition should be the first and most important priority for athletes. Athletes can use body weight strength training as well as plyometrics to improve all sides of their game. The body weight training uses the fundamental actions of moving back and forth in opposition to the force of gravity. In addition to body weight, the gravitational forces experienced during these exercises can act as additional resistance during training. This sort of body weight training can enhance the power capacity of athletes. Other advantages of this training are that this training can develop new neural connections and

promote hypertrophy of muscles. Because of this, using body weight exercises helps improve an athlete's running and rapid change of direction. (Zulkarnain et al., 2021)

Enhancing agility performance to the utmost level has always been the focus of athletes and their coaches. Some studies have revealed that following a training protocol comprising of conventional lower-body resistance exercises, such as Olympic squats and deadlifts, showed little to no improvement in athlete's agility. However, many researches that appraised the effect of strength training in combination with comprehensive agility training have shown more positive results. Therefore, doing squat jumps at higher intensities and with loads greater than which players are familiar with can improve the athlete's change of direction performance. (Human Kinematics)

Explosiveness is a vital characteristic for athletes especially for those who want to have improved vertical jump performance. Therefore, explosive strength is very important for athlete to develop high level of strength and force that can help them generate enough propelling force required during jumping. Strength training, plyometric training and following the right technique for jumping are the ways by which one's vertical jump performance can be improved. Athletes, by enhancing their explosiveness, can excel in any sport. So, squat jump, a form of plyometric training, can not only improve the power of lower limb muscles but also enhance the explosive extension of the hip, which is the basis of athletic performance. ("vertimax,")

During most sports athletes need to employ larger forces within a small duration of time during push-off and landing, that's why strength plays an important role in athletic performance. Reported peak values of vertical ground reaction forces vary during different phases of athletic action. Such as during push-off ground reaction forces can be more than twice the athletes body weight but during landing it can escalate up to four to six times the athletes body weight. To make players competent enough to tackle these increasing ground reaction forces, improving maximum strength, especially high-velocity strength, and power are of major importance during training players. (Keiner et al., 2022)

Explosive power produces force at a very fast rate and is considered very important for its role in improving performance. There are various training methods in advance sport training aimed at increasing explosive power such as common explosive workouts and

those that target specific fitness. Weightlifting, supporting your body weight, doing jumping exercises and making deep jumps are other ways to train. The study supported that use of combination of varying strength training regimes helped to improve physical fitness. (Yang, 2023)

The explosion force is the neuromuscular capacity to overcome resistance with the highest contraction speed. It is made possible by the joint effect of speed and power. This kinetic combination of speed and power is necessary for improving the ability of the legs to generate a lot of force. Doing squat jumps is another method in plyometric exercises that targets explosive power, which is essential for generating the force needed during the bowling action. Plyometric exercises such as squat jumps directly affect the lower limb explosive power, which is necessary for the athlete who requires rapid acceleration and deceleration for better performance. (Hou & Li, 2022)

The indispensable quality of cricketers is having optimal lower limb strength and power. This is true as they support being swift and accurate which are skills needed for fielding and wicket keeping between the wickets. To dismiss the ball at high speed and absorb ground reaction forces generated during balling action high level of lower limb strength and power is required. Another distinguishable feature of cricket is the delivery of repetitive, sporadic or quick movements of medium to high intensity during a shorter duration. To do these movements of varying intensities, athletes should have a high level of both aerobic and anaerobic capacity. (Ghosalkar et al., 2020)

Motor fitness is the ability of a person's neuromuscular system to do well in specific skills that include agility, balance, power, coordination and reaction time. Motor fitness is a crucial component of cricket as it influences players' bowling, batting and fielding action. Enhancement in athletes' speed, power, and endurance during a match can be obtained by good motor fitness. These improvements can result in upgraded sports action on field and lower injury rates. Good motor fitness of cricket players results from versatile training program that includes exercise regimes used to optimize power, agility, and speed. (Boby, 2023b)

Plyometric training combined with resistance training improves motor fitness. Plyometric training has a quick loading period and this is followed by the explosive, powerful

contraction. During this, the synergistic muscles are activated through the stretch-shortening cycle. Plyometric involves the three phases: stretching or eccentric muscle activity, amortization, and concentric muscle action. These phases collectively called stretch shorten cycle of muscle action and help in improving the agility, stability, balance, speed of athletes aiding in increasing speed speed-strength ratio, which generates more force during quick start and faster propelling off the ground. (Dave & Singh, 2024)

Plyometric training is designed to increase the rate at which your muscles contract, your brain's reactions and your speed and agility. Plyometric training works by using the stretch-shortening cycle. The action uses concentric force to build energy and then eccentric force to keep stores elevated; by adding the stretch reflex, energy is released instantly and explosively, giving power and strength. It provides benefits in sports strength, jumping and keeps injuries at bay. (Huang et al., 2023)

In dynamic sports, working on jumping, hopping, skipping and bounding improves the ability to move muscles quickly through a type of training called plyometrics. Most athletes perform plyometric training in dynamic sports to boost their strength and how explosive they move. In plyometrics, you first lengthen a muscle quickly and then pay attention to quickly shortening the same muscle. With periodic strength training, plyometrics has been demonstrated to improve one's explosive jumping skills. In addition, exercises that involve jumping and landing or plyometrics, seem to help build muscle strength and coordination in the lower part of the body better.(Chandra et al., 2023)

Initially, Verkhoshanski divided plyometric training into impact (in which extrinsic weight is used) and non-impact (in which no extrinsic load is placed) types. But modern studies have categorized plyometric training into three types. The first one is traditional (which includes drop jumps and standing jumps), assisted (when help is provided by stretchable bands) is the second one, and the last one is resisted (in which exercise is done in different situations like underwater or weights). Several experimental studies have reported that the physical capabilities of athletes which including muscle strength, explosive power, and running time, have been improved by giving them a regimen that includes plyometric training.(Kons et al., 2023)

Plyometric training has been proposed for the transfer of explosive-power performance and especially for the betterment of vertical jump ability. Neuromuscular strength and efficiency are considered important components for a successful athletic performance. Different training programs involving weight training, explosive power training ballistic training are used for the enhancement of strength and performance. Strength training involving plyometric training for the lower body application includes jumping, hopping, and bounding. Plyometric exercises are countermovement jump, drop jump, and squat jump. These exercises can be performed with a combination of other training programs or may be applied as an independent regimen. (Bugti et al.)

Squat jump, a form of plyometric exercise, increases athletic performance by targeting explosive power. Squat jump increases the peak performance. Targeting the peak & average power, peak & average velocity, and maximal jump performance, the squat jump is considered as one of the best plyometric exercises. Irrespective of morphological change in cross-sectional area or muscle fiber type, improvements in functional performance indicate sarcomere ultrastructural adaptation. The sarcomere ultrastructural adaptation helps in improving the peak performance of players. (Isaacs et al., 2022)

Effective training allows athletes to improve their motor and technical skills which usually leads to physical improvements and making the player stronger. According to Lenhert and his colleagues, having extra strength is connected to better-speed force-related qualities (e.g., force rapidness and power-generation capacity) which are essential for sprinting, changing directions, jumping and special sports skills. (Kabacinski et al., 2022)

Squat jump and plyometric jumps are fundamental exercises that are used not only to obtain explosive power in sports but also the optimal level of agility. These exercises can be done individually or in combination with other exercise models like high-intensity cardio training. These exercises engage the musculature of lower body like the abdominals, gluteus, hamstrings and lower back, and strengthen these muscles effectively. Many studies revealed that squat jump can optimize athletes sprinting capacity as both uses muscle strength and explosive power. That's why many coaches add these exercises in the training regimes of athletes who want to enhance their sprinting speed, explosive power, and agility. ("very well fit,")

Strength-developing drills are pivotal in the training of players after an injury. Explosive Strength training is crucial in a lot of sports and act as a foundation for complex and specialized technical actions. As in most of the group sports (like basketball and hockey), athletes need to move quickly and with high velocity, which is why explosive strength is a critical requirement of the physical abilities of such athletes. Many studies have revealed that for athletes who want to optimize their power, resistance, and speed-velocity profile, strength training plays a pivotal role. Therefore, in such high-intensity group sports to improve the sprinting speed generation of explosive strength. (Melián Ortiz et al., 2021)

A key strength exercise that many use is practicing a vertical jump. It is looked upon as a key part of strength and conditioning training programs. Athletes must produce a lot of force and power to carry out various skills rapidly like accelerating, sprinting, jumping, stopping and turning. Naturally, being very strong and powerful is considered key for success in most sports. Widely, coaches use vertical jump tests to keep track of an athlete's power and improvement. (Fu et al., 2023)

Athletes have vertical jumping measured frequently as part of their regular sports monitoring program. The squat jump (SJ), countermovement jump (CMJ) and drop/depth jump are the most widely studied vertical jump types. Experiments often evaluate the jump height, based on net impulse and the way forces are used. Thus, if you jump with a big force, you could reach the same height in less time as you would with a smaller force and more time. Many vertical jump trainers find resistance training (in the form of weight lifting, power training and plyometrics) to be the most successful method for better performance. (Suchomel et al., 2020)

Agility, being the main component of the performance, can be successfully developed if training is based on changing direction of movement quickly. The agility T test among various agility tests is an appropriate for estimating agility.(Baja et al., 2022) The 30-meter sprint test is a reliable and valid test for testing speed, which is a part of the physical fitness components (involving agility, speed, and power) (Ab Rahman et al., 2021). The vertical jump test for determining the jump height is a cost-effective, reliable, and valid method of assessment. (Comyns et al., 2023)

Jumping involves your arms, legs and the connections between them. It helps show whether a person is healthy and fit for activity in everyday life and in various sports. The momentum of the vertical jump in the thrust phase comes from the strong, explosive actions of the lower limbs. It is the most popular way to test how explosive the power of the lower limbs is. The goal of jumping tests is to give clues on explosive strength, how well muscles coordinate and their ability to react quickly. (Di Domenico et al., 2024)

Basically, physical fitness tests are planned to meet one or more important objectives. Many tests are used to check agility, speed and several other parameters. Agility means being able to change directions quickly and is important in sports, including badminton, tennis, trampoline, gymnastics, hockey, football and basketball etc. Speed is about the speed and control of specific actions and how fast someone can do repetitive movements. Speed refers to one's skill in quickly doing the same movement pattern over and over. (Kumar et al.)

The agility test is commonly used to see how well players can perform changes of direction in various movement planes. The movements for this include sprinting, shuffling side to side and moving backward in several 4-direction changes. The movements in the agility T-test are used in various sports, still, hand-eye coordination exercises in basketball also use an altered version of the original T-test. Because basketball movements are fast and go in multiple directions, the modified version has reduced the total distance of the T-test. (Scanlan et al., 2021)

CHAPTER # 2
LITERATURE REVIEW

Literature Review:

An experiment assessed how using different coaching techniques and training plans can affect the bowling pace of players in amateur cricket. Focusing on those who bowl in cricket at a medium-fast speed, the research considered how regular power training and resistance influenced their bowling speed. Researchers included twenty medium-fast bowlers and checked their bowling speeds both before and after a 2-month strength and power training session. Focusing on creating explosive strength and fast movements, the training was designed with moderate to high intensity. According to the findings, power and resistance exercises helped the sports performance of the cricket players. (Khalid et al., 2025)

M Sun and K.G. Soh conducted a systematic review and meta-analysis on athlete performance due to speed, agility and quickness training in 2025. For this systematic review, eleven RCTs involving 499 participants were studied. The analysis involved examining in detail how SAQ training affects sports performance. There was improvement in the 5-meter and 30-meter sprint tests during the study. It was found that SAQ training improves sprinting, change of direction skills, reaction time, lower body strength and flexibility to a moderate or small degree. (Sun et al., 2025)

Tic-Tac-toe exercises on basketball players were studied in 2024 by Sitanggang, D.L.F., to help improve their agility, speed and accuracy. ADDIE (Analysis, Design, Development, Implementation, Evaluation) was the main structure used for development in this research. The model was checked and approved by the three expert reviewers. Valid and reliable tests were applied using instruments and paired T-test proved that tic-tac-toe exercise improves speed, agility and accuracy. (Sitanggang, 2024)

A study was conducted by Ravi and Diviya in 2024 on the science of plyometrics. The purpose of the study was determining the relation between agility and athletic performance. The experimental methodology employed in this study used factorial analysis and a population of 40 athletes selected using ordinal pairing. The post-test indicated a significant value, indicating that with the following of exercise regimens, there was an enhancement

in agility. The Illinois test was used as the post-test. With the enhancement in agility, there was an overall positive effect on athletic performance. (Ravi & Divya, 2024)

Both Aryadi and Ayadi reviewed the effects of squat jumps, also regarded as jump to the box, on improving jump power and performance. Participants were assigned to two groups and each group participated in a pretest and a posttest. The data analysis included a hypothesis test (t-test) to assess whether there was a difference between pretest and posttest, using vertical jumps to measure the results. According to the research, squat jumps make athletes more explosive and strong in their lower limbs. The average score after the test was greater than the average before the test with a dramatic difference. (Aryadi & Ayadi, 2024)

Williams et al. in 2024 carried out a cross-sectional study to find out how fast youth basketball players are able to accelerate during the Tic-Tac parkour action, drop jump and lay up shot exercises. Results showed that doing the parkour Tic-Tac style can improve an athlete's skills and speed more than both drop jump and lay-up approaches. (Mark David Williams et al., 2024)

A 2024 study investigates how training with ballistic-squat jumps and only weights (with no slowing down) affects explosive power. Twenty athletes who were well trained were assigned by chance into experimental and control group. The experimental group wore wrist weights for squat jumps at a weight equal to 30% of their 1 repetition maximum and the control group completed jumps with no extra weight. Results revealed that both groups got stronger in explosive movements. (Yi et al., 2024)

A 2024 study by B. Turkarsalan and his group. The study was designed to find out how doing plyometric exercises influences a young soccer player's speed, agility and ability to jump. There were 26 players who were involved in training as part of their sports club. The results of the statistical analysis showed that doing plyometric training improved agility, increased the vertical jump and enhanced 30m sprint speed. (Türkarslan & Deliceoglu, 2024)

SJ Callaghan and Co researched how strength training influences the contact with the ground at the front foot and the force used to release the ball when bowling in elite cricket

players. The impact a training program lasting eight weeks has on lower-body strength, power, eccentric ability, the foot position during hitting and the resultant ball speed was analyzed with pre- and post-tests. Changes between the two scores were checked using paired sample t tests. The study found that doing strength training can strengthen both the fast and precise movements of pace bowlers. (Callaghan et al., 2024)

E. Abbas and A. Zahra studied how Multimodal High-intensity Interval Training influenced the speed, agility and overall performance level among cricket players, in research done in 2024. Therefore, this study was conducted to figure out how much this method improves the lower limb exercise abilities of elite cricketers. People from the Pakistan cricket board and cricketing academies of Lahore were taken and tested in a randomized control study. This study found that multimodal training improved results more than did just classic training. (Abbas et al., 2024)

In 2024, Sneha Jawalker and Co performed a study to study the effect of combining plyometric training with dynamic stretching on how fast male cricketers can run. Two groups were made from a pool of forty male cricketers by using random selection. It was found that after 2 weeks of plyometric training and dynamic stretching, the speed of cricketers belonging to group A improved more significantly than those belonging to group B who played conventional training. (Jawalkar et al., 2024)

A study on the topic The relationship between agility, linear sprinting, and vertical jumping performance in U-14 and professional senior team sports players was conducted by T. Gisladottir, M. Petrović in 2024. After participating in different games these players were then evaluated using different tests, such as modified t-test for agility, 20-meter sprint test for speed, and counter jump for explosive power. After measuring the results by using Pearsons product-moment correlation coefficient, this study concluded that linear speed and explosive power have a positive effect on agility and change of direction speed. (Gisladottir et al., 2024)

A meta-analysis was done by L. Liu, X. Niu, and Z. Zhou in 2024. This systematic review investigated the improvement in jump and sprint performance by the implementation of resistance training, plyometric training, and mixed training. To obtain the data for this systematic review, articles were searched through different databases like Web of Science

and PubMed. The result of this analysis demonstrated that significant improvements were seen with the implementation of resistance training modules, and no significant changes were observed by using plyometric or mixed exercises.(Liu et al., 2024)

In 2023, H. Haung and WY Haung explored how speed, agility and explosive strength were improved in elite athletes through plyometric training. Fifteen men from college basketball teams were given plyometric training designed for elite athletes three times per week for 8 weeks. Before and after taking the course, students had body composition, speed and agility and explosive strength tests. Once training was finished, people showed lower body mass index and body fat percentages, a rise in the skeletal muscle mass and faster and better outcomes for speed, agility and explosive power. (Huang et al., 2023)

In 2023, Latif and team conducted research about how many plyometric exercises on agility influenced cricket fast bowlers. To begin, assessments were taken by running the Illinois Agility Run Test, T agility test and Vertical Jump Test. The assessments were then done again on the third and sixth weeks. The analysis of tests demonstrates that the experimental group has a major effect. After the study, it was found that plyometric and agility training (such as tic tac toe training) enhanced the agility of cricket players. (Latif et al., 2023).

Forster and coworkers examined available evidence using a systematic approach to find out how training can advance pro-agility performance. Twenty studies were picked and we distinguished and analyzed 29 groups that tried out various training programs for their primary outcomes. Sprint, plyometric, resistance and combined resistance training were found to increase pro-agility test results in every training session.. (Forster et al., 2023)

A systemic review done by X Wang, CLv in 2023 studied how the plyometric training related to the explosive power and strength of lower limb. Healthy athletes and sports students with previous training experience were included in this study. After giving the intervention for 6-weeks jump ability, sprint ability, and 1-RM were calculated to evaluate the result. Jump ability was tested by squat jumps, sprint ability was tested by the 10-meter sprint test, and strength was tested by partial or full squats. This study concluded that no

significant difference in explosive power by both the intervention (plyometric training and complex training).(Wang et al., 2023)

Research indicates that doing squat jumps improves the lower body's ability to move fast which is needed for various games that call for quick starts and jumps upwards. Isaac et al. found that even inactive people who followed an eight-week plyometric squat jump training program improved their mobility scores and vertical jump height and speed. No visible muscle growth or transformation in muscle fiber types was observed, but the researchers think improvements in performance could be connected to sarcomere changes.(Isaacs et al., 2022)

In 2022, D Abi Permana and other researchers examined a six-week plyometric program given to professional athletes in East Java. The researchers wanted to know if plyometric training made athletes faster and more powerful. The outcomes revealed that athletes became more explosive and agile from before to after the program. The results proved that the athletes developed more explosiveness and agility after training.. (Abi Permana et al., 2022)

SE Luther and Co, conducted a study on how frequently doing core and agility activities affected recreational cricketers' sense of balance. It was a study that used a single group and a pre-test-post-test design. A Y-balance test was used for the pre-test post-test analysis. The main parts of the intervention were the warm up, the training and the cool down phases. A paired t-test was used to find out how much the scores changed in the pre-test and post-test measures. Training on core strengthening and agility made participants significantly better at core strength, balance and agility. (Luther et al., 2022)

A study was conducted in 2022 aimed to find out the comparison in the effects of technical weightlifting and plyometric exercise on vertical jumps as well as on the sprint performance. 45 male participants took part in this study, all of whom underwent a 4-week learning period focused on weightlifting derivatives. After initial testing was conducted, and participants were then randomly allocated into groups. Following this, the participants followed an 8-week training program. After post-testing the results concluded that

plyometric training was more beneficial than a weightlifting program in enhancing both unloaded and loaded vertical jumps, as well as sprint performance. (Berton et al., 2022)

S. Kosova and Co. performed a study in 2022 to see how effective plyometric exercises are. The purpose of the study was to check the impact of 8 weeks of plyometric exercises on the jumping skills, speed and agility of teenage fencers. For the study, 26 participants were put into groups by a random process. For one group, plyometric training was included in their plan and the other group just practiced fencing. There were positive changes seen in jump performance, speed, and agility. (Kosova et al., 2022)

Last year, AK Ramachandran and his team did an in-depth review and analysis of how strength and conditioning supports the improvement of speed and accuracy for cricket fast bowlers. This study used a systematic review and meta-analysis to check how different training programs change bowling speed and accuracy. Articles were included from PubMed, Web of Science, SPORTDiscus, Scopus and CINAHL. The study revealed that fast bowlers gained a lot from the plyometric training program. (Ramachandran et al., 2022)

In 2022, CH Chuang and MH Huang looked at the impact of agility training on young female volleyball players' physical skills. Thirty female players were placed in each of the three groups: shuttle run training, agility ladder training and control, using randomization. Evaluations were made with agility t-test, 10-meter sprint and digging agility test before and after training. It was determined after analysis that the those who did sprint and agility training performed better than the control group. Therefore, the research found that doing agility training enables sprinters to move faster in side-to-side sprints.(Chuang et al., 2022)

A study undertaken by Jiang and Xu examined what happened to the explosive power of young basketball players' legs after doing chain squats with four different chain load ratios. Screening took place after 6 weeks of training with the training plan of two interventions a week. Tests were administered using a 5-meter shuttle run, a 30-meter sprint test, standing long jumps and a one-repetition maximum (1-RM) squat. It was found that adding more chains to lower body squatting exercise improves explosive power. (Jiang & Xu, 2022)

A group of researchers did a study in 2021 to understand how plyometric training and resistance training affect the speed and power of leg muscles. The team used a pre- and post-tests in a sample consisting of 60 students. Data was collected through 30 meter run test and standing broad jump test which were conducted using a purposive random sampling design. Mixed-model analysis of variance (MANOVA) was the analysis technique employed. The findings demonstrated the p-value to be less than the significant value of 0.05 which showed that speed and explosive power improved due to training. (Arianda et al., 2021)

In the year 2021, Priya and Umananda studied the link between explosive strength and agility. Of the twenty participants, all male college athletes, every person took part in both countermovement jump and t-test tests. It was shown that having strong explosive strength is linked to being less agile. Stronger explosive power means the time taken to finish the agility T-test is reduced. Improving explosiveness by doing different drills leads to better agility. (Priya et al., 2021)

Research was conducted in 2021 by Marker and Taliep to see the effect of resistance training on the bowling action of fast bowlers. Eighteen players were included in this study, which was then divided into a combined resistance training group (CRT) and regular training group (RTR). A four-week training session was given to both groups, in which the CRT group was given a protocol involving core strength training with plyometric exercises, and the RTR group followed the conventional training method of balling with a weighted ball. After analysis of the results, no significant difference between the effects of the two training sessions was seen. (Marker & Taliep, 2021)

A research project was performed by AM Ortiz and DV Delgado in 2021, to see how different exercises for the hamstrings and quadriceps impact explosive strength. For this RCT, 80 volunteers were sorted into four groups through single blinding. All four groups used concentric, eccentric, concentric-eccentric, and isometric exercises. Weight, height BMI, heart rate, jumping ability and 60-meter sprint speed were the central variables in this study. This research found that enhancements in the main muscle-related variables occurred after doing concentric and eccentric exercises. (Ortiz et al., 2021)

In 2021, researchers looked at the effect of doing body weights and plyometric exercises on athletes' speed and agility in taekwondo. Purposive sampling was adopted to gather the data which was afterwards split into 2 groups. Students ran a 30-meter sprint test and did the Illinois test of agility as a pretest. This study which lasted 6 weeks, was analyzed with the MONOVA test approach. The study indicated that plyometric exercises give taekwondo players better speed and agility than other types of workouts. (Zulkarnain et al., 2021)

Elite cricketers were investigated by Herridge and Co over 20 weeks off-season to examine how their power, speed, agility and endurance progressed. During the off-season period, fourteen top male cricketers did a physical testing battery in both week 1 and week 18. A squat jump was included among the tests and they showed these tests had a significant effect on power, speed, agility and endurance. (Herridge et al., 2020)

The way you do the exercises and train also affects how effective squat jumps are. Garcia et al found that doing jumping exercises in a deeper squat helped achieve better results and greater peak power. All the participants in the study were 13 resistance-trained males who qualified according to the inclusion criteria. Included in the criteria were that players had no injuries in the past years, had a minimum of 2 years of experience training and participants were asked not to consume caffeine for 24 hours before workouts. (Garcia et al., 2020)

L Villalon-Gasch conducted a study in 2020 demonstrated that there is a correlation for the enhancement of vertical jump height and explosive power given with squat-based exercise program. The study involved thirteen trained volleyball players randomly divided into two groups, one group given the squat-based exercise and the other group continued the previous regimen. The end result showed that squat based exercise improved the vertical jump height and performance level while the other didn't show improvement. (Villalon-Gasch et al., 2020)

An experimental study design was used in 2020 to examine through which training program physical fitness components of athletes improve. In this study 40 male students were assigned into three groups through stratified and proportional simple random sampling. One group was assigned as control followed no training while other two were assigned as experimental groups, followed by plyometric and resistance training. Data was

collected with pre- and post-test involving a 30-meter sprint test, a T-test, and vertical jump test. Plyometric training was seen to have had greater increase in speed, agility, and explosive power. (ABEBE, 2020)

In 2020 research was done to see how change-of-direction speed could be improved in junior team sport athletes. This study concentrated on observing which training model led to the biggest improvement in change-of-direction speed by testing horizontal load training, vertical jump training, and explosive power training on 45 male athletes. Evaluation for the pretest and posttest of 1-leg lateral jumps, standing long jumps and drop jump heights was carried out. They saw an increase in how fast people could change direction in the four groups. (Keller et al., 2020)

Being a fast bowler in cricket means being prepared for high performance. Sai Aditya Krishna and his Co performed a 2019 study to assess the influence of plyometrics on knee strength and jump ability in cricket fast bowlers. Forty-two bowlers were randomly picked to be in the training group and another group that did not train. They all did tests to measure their isokinetic knee power and jumping abilities. Overall performance of the fast bowlers was better and this training improved their knee strength. (Krishna et al., 2019)

A group of researchers, led by Ferreira, examined squat jump performance and the role of power, agility and speed in 2019. Spearman correlation coefficient was used to look for connections between variables. Among the measures used, lower limb power proved a powerful establishment that helped predict a player's ability to skate and turn sharply. According to the research, there is a strong link between the power, agility and speed. (Ferreira et al., 2019)

Iris Dijkma and her team carried out a cluster randomized pilot study in 2019. Content was about how agility training can change body control, the ability to change direction, speed and injury rates in Dutch recruits. Throughout the 23 weeks, 64 recruits took part in the training. Recruits took part in testing of body control (using the agility t-test) and change of direction speed (with the Illinois test) before and after the training sessions. It was shown that agility training might help in controlling one's body movements, changing directions and preventing injuries. (Dijkma et al., 2019)

In 2018, a randomized controlled trial investigated how lower-limb strength and speed were changed by combining strength training with plyometric exercises. Half of the participants were randomly chosen to do plyometric exercise for 8 weeks and the rest continued following their usual workout routine. Everyone in the study did the 20-meter sprint test and squat jump to determine power, velocity, force and height. Sprint test and squat jumps scores improved more for the experimental group than for the control group. (Fischetti et al., 2018)

Coratella and Co tested adapted performance and muscle build by training with weighted squats and body mass squats in soccer players. The results suggest that playing with external loads can improve athletic results related to change of direction, sprinting or jumping. Sprinting, changing direction and jumping seem to benefit more from weighted squat-jump than from body mass squat jump. (Coratella et al., 2018)

B. Bashir, F. Suleh Hayyat and S. Shafi conducted an experiment in 2018 to check the influence of plyometric and resistance exercises on cricket players' agility, speed and explosive power. When the training protocol was done, the data was further analyzed with the paired t-test, independent t-test and covariance. It was proven by the study that performing plyometric training made people move faster, be more agile and increase their explosive strength. Thus, researchers found that plyometric exercises help cricket players improve their sports skills more than agility training. (Bashir et al., 2018)

In 2017, a study by Loturco looked at the effect of doing squat jumps. The purpose of the study was to investigate whether achieving greater maximum mean propulsive power (MPP) in loaded squat jumps positively affected overall athletic performance. Twenty-two women made the team. Training with squat jumps improved sprinting, change of direction speed and jumping skills more than training with higher squats. According to the study, doing squat jumps as part of a training regimen raised participants' speed-power performance. (Loturco et al., 2017)

Shakow, M. did an investigative study to examine how mental abilities are related to determining accurate second moves in tic-tac-toe by trail making. Scoring better on the Trail-Making test is linked to choosing the proper prompted second action in Tic-Tac-Toe,

the study showed. It can be seen from the results of a T-test that those who did better in the Trail-Making test tended to play a more accurate second move in Tic-Tac-Toe. (Shakow)

A researcher called Yakup Akif Afyon carried out an RCT in 2017 to see how core training affects speed and agility in soccer players. He included 40 amateur soccer players who chose to take part in the study and divided them into two groups. For 4 days each week, the athletes went through an 8-week training routine. Before and after training, the athletes were required to do a 30-meter sprint for speed and took the Illinois test and t-test for agility. Data from this study were tested with a paired t-test. Results showed no significant change from the initial testing to the final testing. (Afyon et al., 2017)

Research was done by WJ Style and MJ Matthew on effects of strength training and sprint performance in soccer players. The purpose of the study was to check whether maximal squat strength and sprint action in elite players increase with basic in-season strength training. Players were evaluated before and after a 6-week training session by checking their height, body mass, and 1 RM for squat and sprinting speed (by 5, 10, and 20 m test). The results showed apparent changes in absolute and relative squat strength after strength training. As well as an enhancement in sprint speed also occurred after this training. (Styles et al., 2016)

In 2016, Marián and his colleagues performed research to learn what the effects of 8-week squat jump training were on Fmax, rate of force development, CMJ and SJ heights, and the sprint of untrained individuals. In the study, participants were sorted into an experimental and control group, which were tested at the beginning, mid and end points of training. Experimental participants had a significantly higher increase in Fmax. A decrease in SJ and CMJ height, as well as improvement in 50-meter sprint time was observed in the group given the training program. (Marián et al., 2016)

In 2015, Loturco and his colleagues studied the effect of mean propulsive velocity that is determined by the Power Load in Jump Squat. Using the Mean Propulsive Velocity. This study checked if the same height for jumps can be achieved by professional players from different sports, even if these sports have varying maximum mean propulsive velocities.

109 elite players from varying sports were included in this study, who were given progressive loading starting from 40% of their body mass. Results stated that maximum mean propulsive power and velocity were achieved regardless of the sport from which the athlete was taken. (Loturco et al., 2015)

CHAPTER # 3
RATIONALE OF THE STUDY
AIMS AND OBJECTIVES
HYPOTHESIS

3.1 Rationale of the study:

Fast bowling in cricket is a multi-articular action performed at high speed. Fast bowlers require a unique fusion of physical attributes, including speed, agility, and explosive power, to excel in their performance. Their ultimate goals are to produce maximum force in a limited time by enhancing explosive power, reducing the risks of injury through agility training, and increasing the speed, which ultimately raises the velocity of the bowler, which affects dismissals.

Depending upon the effects of the training program, we can recommend clubs and cricket academies incorporate tic-tac-toe drills and squat jumps into their regimen.

3.2 Aims and objectives:

The aims of the study were the following:

- To increase speed and agility
- To increase explosive power
- To improve overall performance in sports

3.3 Hypothesis:

3.3.1 Null Hypothesis (H^0):

There was no significant difference in the comparative effects of Tic-Tac-toe exercise and squats jumps on the agility, speed, and explosive power among the fast bowlers.

3.3.2 Alternative Hypothesis (H^A):

There was a significant difference in the comparative effects of Tic-tac-toe exercise and squats jumps on the agility, speed, and explosive power among fast bowlers.

CHAPTER # 4
MATERIAL AND METHODOLOGY

Material and Methodology:

4.1 Study Design:

This study was a randomized clinical trial (RCT).

4.2 Sample Size:

The sample size was 42. (Krishna et al., 2019)

4.3 Sampling Technique:

Purposive sampling was used to select participants.

4.4 Study setting:

- Jawad Club
- Imran Ali Cricket Academy
- Saeed Ajmal Cricket Academy

4.5 Selection criteria:

4.5.1 Inclusion criteria:

- Age: 18-30 years
- Male
- Fast bowlers
- At least one year of experience.(Oranchuk et al., 2019)

4.5.2 Exclusion criteria:

- Any systemic illness
- History of any sprain or strain (in the last 3 months) (Falces-Prieto et al., 2022)
- History of fracture/traumatic injuries (in the last 6 months) (Jeevannavar et al., 2020)
- History of any neural injury in the last 3 months before performing tests (Falces-Prieto et al., 2022)

4.6 Data Collection Tools:

- Consent form
- Self-made Questionnaire

4.7 Measurement Tools:

- **4.7.1 T-test:** To measure the agility (acceleration, deceleration, and lateral movement), a T-shaped figure was drawn on the floor. Three cones were placed on the ground with a five-meter distance in each two cones on a straight line. A fourth cone was set at 10 meters from the middle cone so that the T-shaped figure formed. The athlete had to move from the center of the T shape forward, side to side, covering a distance of 5 meters towards the right cone, then the player side-stepped to reach the far cone, then came to the middle cone. Finally, the player moved 10 meters backward while the time taken to complete the test was noted with the stopwatch. (Baja et al., 2022)

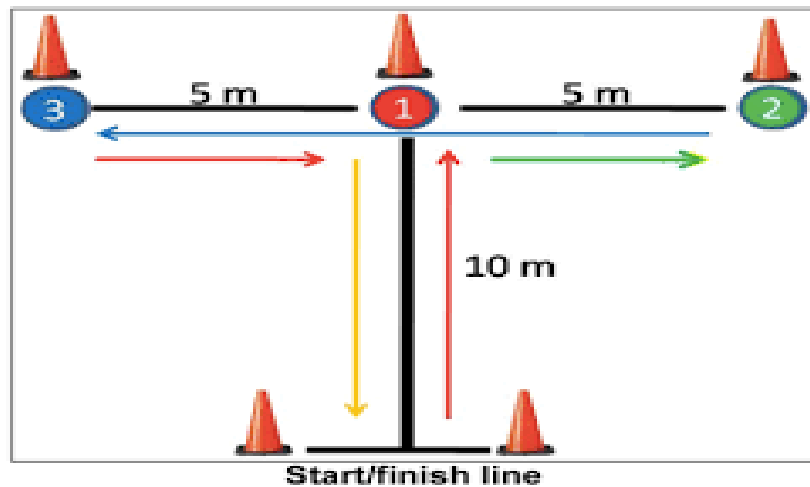


Figure 4.1: Agility T-test

- **4.7.2 30-meter sprint test:** The test is reliable and valid to measure maximum speed and acceleration of the subjects. The test involves a single maximum sprint run of 30 meters, and the time is recorded using a stopwatch. Participants repeat the test twice, and the most appropriate result is recorded. (Ab Rahman et al., 2021)

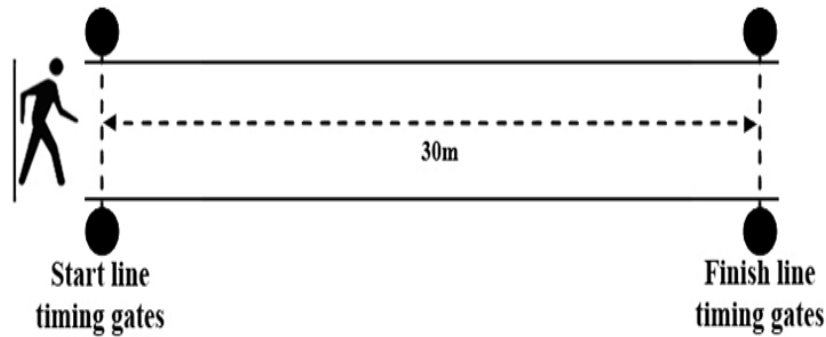


Figure 4.2: 30-meter sprint test

- 4.7.3 Vertical jumps:** Also called as countermovement jumps use to measure the explosive power and the ability of subjects to jump at maximum height. The players adopt the side-on position and raise their arm closest to the wall to its maximum height. The player's fingertips are marked at the feet planted on ground standing. Before initiating the jump, the players bend their knees and then execute the jump. The purpose is to touch the wall and leave a mark on the wall at the highest point achieved during vertical jump. The difference between standing reach height and the jump height is recorded. The best of the three values is recorded. (Comyns et al., 2023)

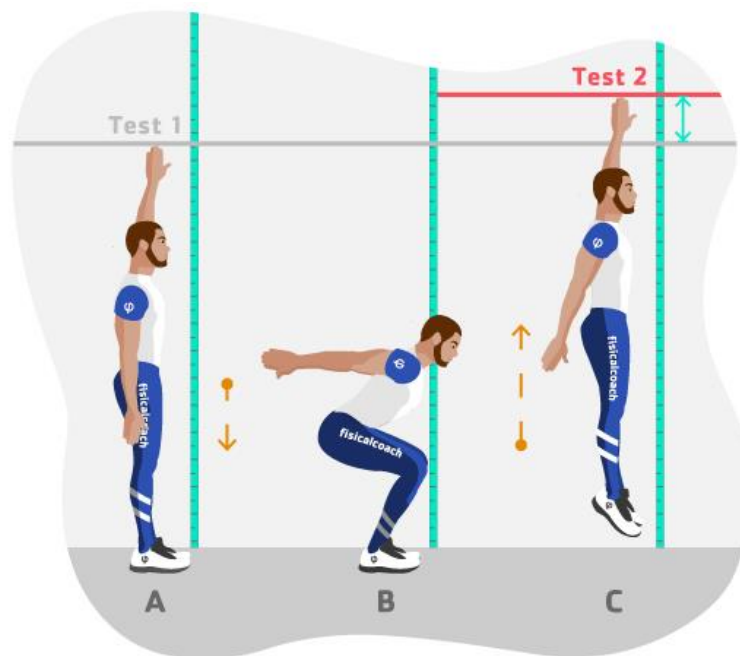


Figure 4.3: Vertical Jump test

4.8 Duration of study:

The duration of study was 3 months after the approval of synopsis.

4.9 Data Collection Procedure:

The study was a Randomized Clinical Trial with a sample size of 42 male fast bowlers aged between 18-30 years. Data was selected on the basis of inclusion criteria with a purposive sampling technique from the clubs and cricket academies of Faisalabad. Before dividing the subjects into two groups initial level of testing was performed by T-test, 30-meter sprint test, and vertical jump test. Coin tossing was used to randomize the subjects into Group A and Group B. Single blinding was used, and Group A followed the Tic-tac-toe drills as an intervention, and Group B followed squat jumps as a second intervention. A total of 6 weeks' protocol was given on alternative days in a week to both groups, with baseline in the form of full-body dynamic stretching. The training session was followed by a cool-down exercise (gentle jogging, breathing exercise). The total training session was 20 minutes, involving 5 minutes of warm-up and 5 minutes of cool-down.

Group A in the first and second weeks followed the same intensity of exercise with 3 reps and 2 sets. The first and second weeks involved simple tic-tac-toe drills, running sideways from the starting point, which was 10 meters away from the tic-tac-toe frame. Then, the players placed the targets on the tic-tac-toe frame.

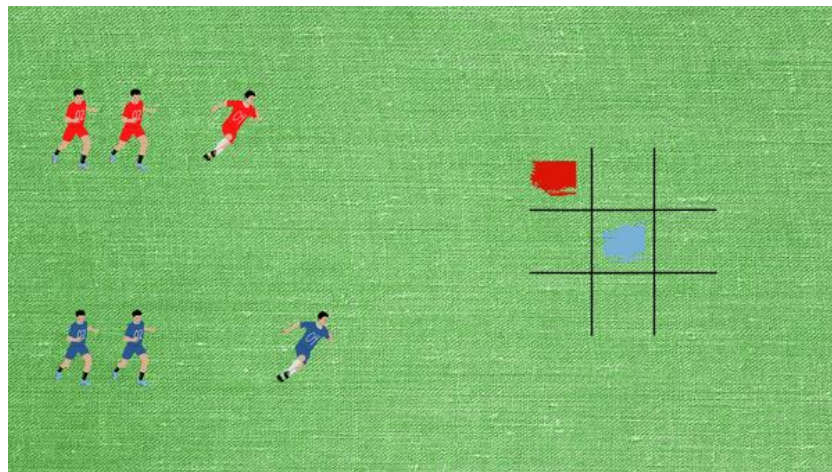


Figure 4.4: Tic-tac-toe drills

In the third and fourth weeks, the intensity was increased up to 4 reps and 2 sets. The tic-tac-toe exercise these weeks involved a somewhat harder level than the previous one, with

the addition of cones in the running trajectory. The subject reached the tic-tac-toe frame with zig-zag movement around the cones and marking their targets.

Fifth- and sixth-week intensity again increased up to 5 reps and 2 sets. The tic-tac-toe exercise was based on running around the cones in a zig-zag manner with the increase in the number of cones and marking the targets. (Sitanggang, 2024)

	Frequency (F)	Intensity (I)	Time (T)	Type (T)
Week 1	3days/week	3reps, 2sets	5-10 min	Tic-Tac-toe drills
Week 2	3days/week	3reps, 2sets	5-10 min	//
Week 3	3days/week	4reps, 2sets	5-10 min	Drills with cones addition
Week 4	3days/week	4reps, 2sets	5-10 min	//
Week 5	3days/week	5reps, 2sets	5-10 min	Drills with the addition of more cones
Week 6	3days/week	5reps, 2sets	5-10 min	//

For squat jumps, a protocol of 6 weeks was given on the alternative days, with every two weeks of the same protocol. The first two weeks of the protocol involved simple squat jumps with 5 reps and 2 sets.

In the third and fourth weeks, the protocol involved 10 reps and 2 sets with directional change in squat jumps.

Fifth- and sixth-week protocol was progressed to weighted squat jumps (0.5 kg on each ankle) with 15 reps and 2 sets. (Coratella et al., 2018)

	Frequency (F)	Intensity (I)	Time (T)	Type (T)
Week 1	3 days/week	5reps, 2sets	5-10 min	Squat jumps
Week 2	3 days/week	5reps, 2sets	5-10 min	//
Week 3	3 days/week	10reps, 2sets	5-10 min	Squat jumps with directional change
Week 4	3 days/week	10reps, 2sets	5-10 min	//
Week 5	3 days/week	15reps, 2sets	5-10 min	Weighted squat jumps

Week 6	3 days/week	15reps, 2sets	5-10 min	//
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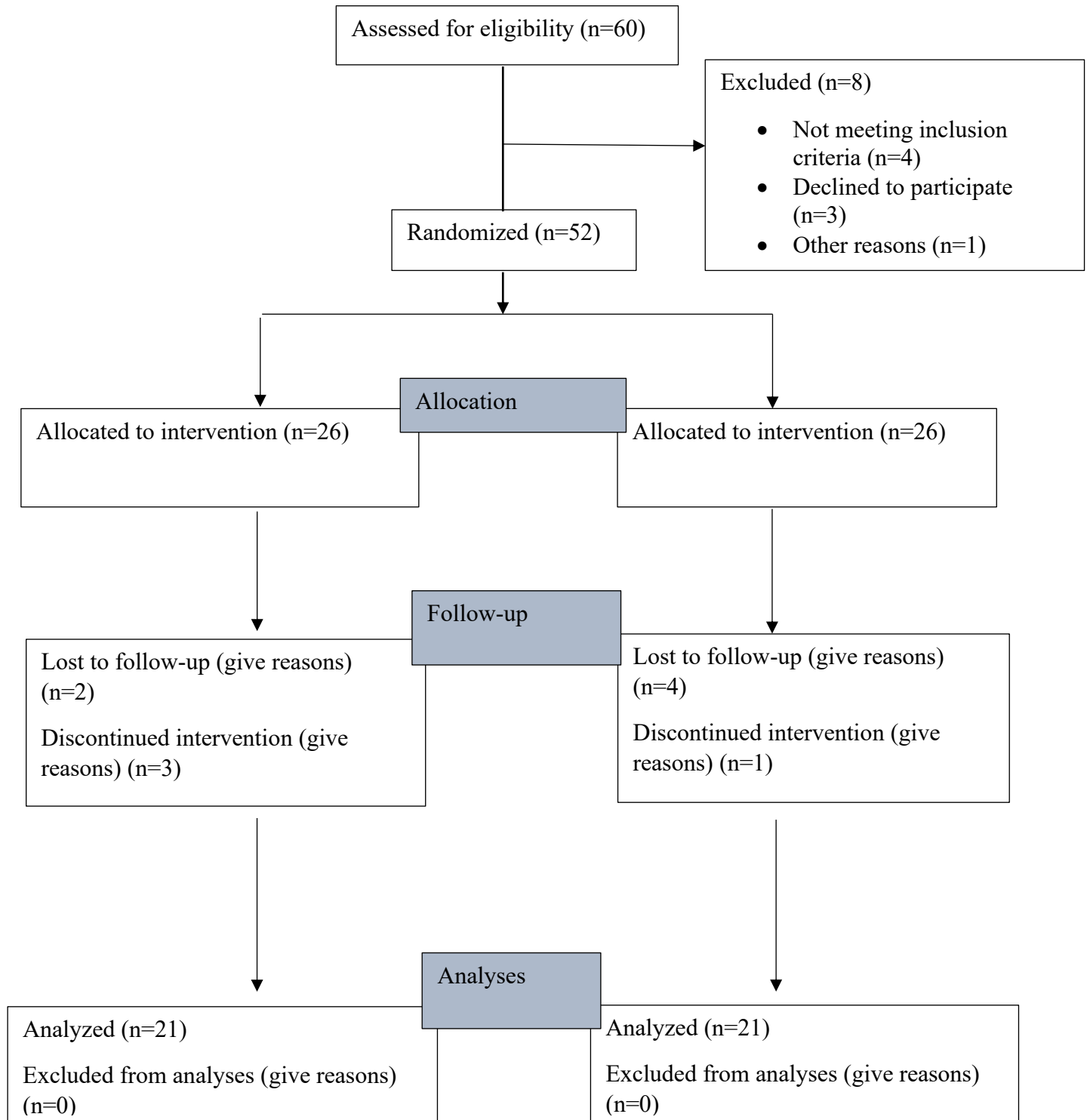
The training session will be followed by a cool-down exercise (dynamic stretching, child's pose, breathing exercise). Also, a warm-up will be given in the form of full-body active stretching. After the 6th week of training, the T agility test, the 30-meter sprint test and the vertical jump test were performed to check the effect of the drills on agility, speed, and explosive power.

4.10 Data Analyses:

The data was analyzed by SPSS-27.

4.11 CONSORT Flow Diagram

Enrollment



CHAPTER # 5

RESULTS

Table 5.1: Group A: age of participants

Group A age category			
Valid		Frequency	Percent
	18-21(years)	6	28.6
	21-26(years)	14	66.7
	26-30(years)	1	4.8
	Total	21	100.0

This table shows the frequency distribution of the age of group A, a total of 21 participants categorized into three groups. 18-21(years) having frequency of 6 with 28.6%, 21-26(years) having frequency of 14 with percentage of 66.7%, and 26-30(years) having frequency of 1 with 4.8%. Overall percentage is 100% with a total of 21 participants.

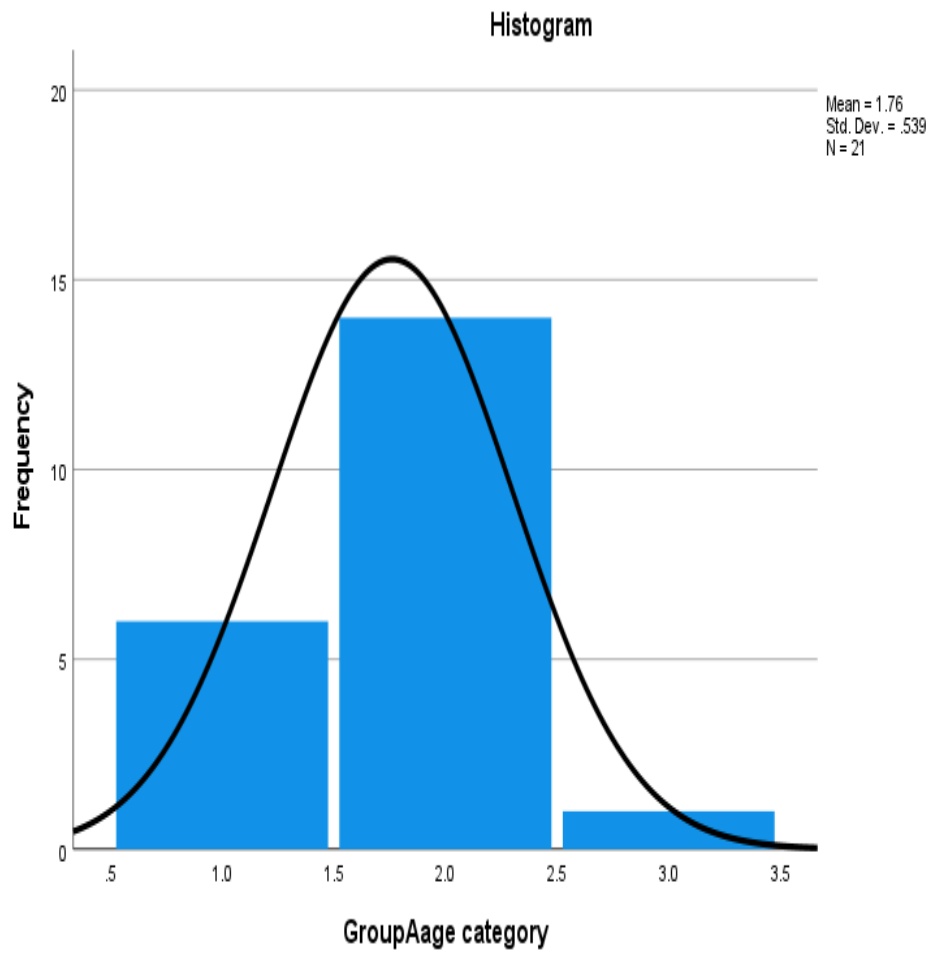


Figure 5.1: Histogram of age of group A

Table 5.2: Group B: age of participants

Group B age category			
Valid		Frequency	Percent
	18-21(years)	6	28.6
	21-26(years)	13	61.9
	26-30(years)	2	9.5
	Total	21	100.0

This group shows the frequency of age of group B, a total of 21 participants categorized into three groups. 18-21(years) having frequency 6 with percentage of 28.6%, 21-26(years) having frequency of 13 with percentage of 61.9%, and 26-30(years) having frequency of 2 with 9.5%. Overall percentage is 100% with a total of 21 participants.

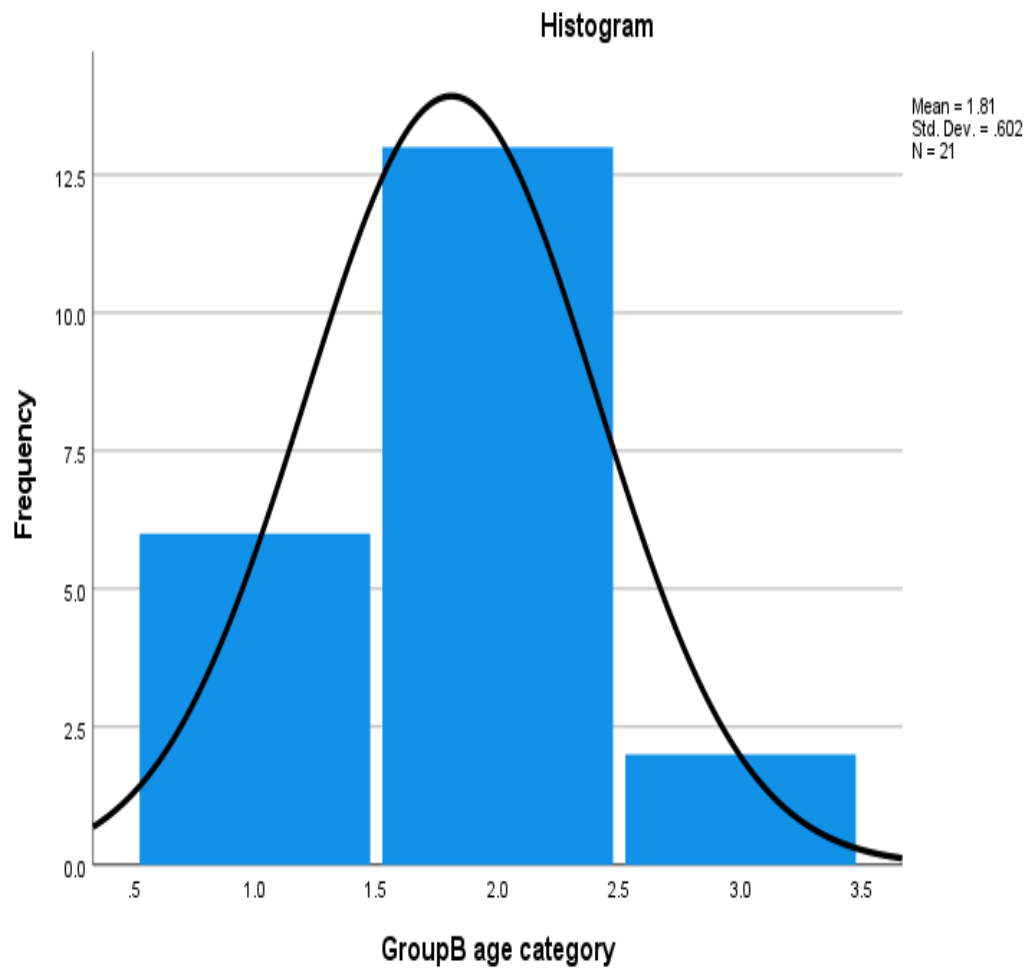


Figure 5.2: Histogram of age of group B

Table 5.3: Frequency distribution of study groups

Groups			
Valid		Frequency	Percent
	Group A (Tic-tac-toe exercise)	21	50.0
	Group B (squat jumps)	21	50.0
	Total	42	100.0

This table shows the frequency distribution of study groups of a total of 42 participants categorized into two groups. Group A Tic-tac-toe has the frequency of 21 with 50%, and group B (squat jumps) has the frequency of 21 with 50%. The overall percentage is 100% with 42 participants.

Table 5.4: Paired t-tests for the 30-meter sprint test

Paired t-tests for the 30-meter sprint test				
Paired sample statistics				
		N	Mean $\pm$ Std. deviation	P value
Pair 1	30-meter sprint test pre-values for group A	21	4.3676 $\pm$.43874	<.001
	30-meter sprint test post-values for group A	21	4.0438 $\pm$.39252	
Pair 2	30-meter sprint test pre-values for group B	21	4.1414 $\pm$.52154	<.001
	30-meter sprint test post-values for group B	21	3.7381 $\pm$.44399	

The table shows that the mean value of pre-values of the 30-meter sprint test for group A and group B were 4.36 and 4.04, respectively. The mean post-values of the 30-meter sprint test for group A and B were 4.04 and 3.73, respectively, after the following the regimen. It means that there was an improvement in the 30-meter sprint test after following both regimens. The results showed the significant effect within the groups since the p values were less than 0.05.

Table5.5: Independent t-test for the 30-meter sprint test

Independent t-test for the 30-meter sprint test				
Independent group statistics				
	Groups	N	Mean $\pm$Std. deviation	P value
30-meter sprint test pre-values	Group A (Tic-tac-toe exercise)	21	4.3676 $\pm$.43874	.136
	Group B (squat jumps)	21	4.1414 $\pm$.52154	
30-meter sprint test post-values	Group A (Tic-tac-toe exercise)	21	4.0438 $\pm$.39252	.023
	Group B (squat jumps)	21	3.7381 $\pm$.44399	

Data analysis of both groups showed a decrease in mean value from pre-test to post-test for the 30-meter sprint test. This showed the effectiveness of both regimens, but the greater decrease in mean value for squat jumps showed that it is more effective than tic-tac-toe drills. This table showed that significant difference between the groups for 30-meter sprint test as the sig.(2-tailed) value was 0.023, which was less than the 0.05 set value.

Table 5.6: Paired t-tests for the Agility T-test

Paired t-tests for the Agility T-test				
Paired sample statistics				
		N	Mean ± Std. deviation	P value
Pair 1	Agility T-test pre-values for group A	21	12.0248 ± .91492	<.001
	Agility T-test post-values for group A	21	9.8648 ± 1.00524	
Pair 2	Agility T-test pre-values for group B	21	12.0729 ± .69037	<.001
	Agility T-test post-values for group B	21	9.9314 ± .09316	

The table shows that the mean value of pre-values of the agility T-test for group A and group B were 12.02 and 12.07, respectively. The mean post-values of the agility T-test for group A and B were 9.86 and 9.93, respectively, after the following the regimen. It means that there was an improvement in the agility T-test after following both regimens.

The results showed the significant effect within the groups since the p values were less than 0.05.

Table 5.7: Independent t-test for the Agility T-test

Independent t-test for the Agility T-test				
Independent group statistics				
	Groups	N	Mean $\pm$ Std. deviation	P value
Agility T-test pre-values	Group A (Tic-tac-toe exercise)	21	12.0248 $\pm$.91492	.848
	Group B (squat jumps)	21	12.0729 $\pm$.69037	
Agility T-test post-values	Group A (Tic-tac-toe exercise)	21	9.8648 $\pm$ 1.00524	.768
	Group B (squat jumps)	21	9.7967 $\pm$.30362	

Data analysis of both groups showed a decrease in mean value from pre-test to post-test for the agility T-test. This showed the effectiveness of both regimens, but the greater decrease in mean value for squat jumps showed that it is more effective than tic-tac-toe drills. This table showed that there was no significant difference between the groups for agility T-test as the sig. (2-tailed) value was 0.768, which was greater than the 0.05 set value.

Table 5.8: Paired t-tests for the Vertical jump test

Paired t-tests for the Vertical jump test				
Paired sample statistics				
		N	Mean $\pm$ Std. deviation	P value
Pair 1	Vertical jump test pre-values for group A	21	14.3476 $\pm$ 2.65812	<.001
	Vertical jump test post-values for group A	21	18.7905 $\pm$ 2.88546	
Pair 2	Vertical jump test pre-values for group B	21	14.0952 $\pm$ 1.60950	<.001
	Vertical jump test post-values for group B	21	20.2048 $\pm$ 1.98884	

The table shows that the mean value of pre-values of the vertical jump test for group A and group B were 14.34 and 14.09, respectively. The mean post-values of the vertical jump sprint test for group A and B were 18.79 and 20.20, respectively, after the following the regimen. It means that there was an improvement in the agility T-test after following both regimens. The results showed the significant effect within the groups since the p values were less than 0.05.

Table 5.9: Independent t-test for the Vertical jump test

Independent t-test for the Vertical jump test				
Independent group statistics				
	Groups	N	Mean $\pm$ Std. deviation	P value
Vertical jump test pre-values	Group A (Tic-tac-toe exercise)	21	14.3476 $\pm$ 2.65812	.814
	Group B (squat jumps)	21	14.1905 $\pm$ 1.47034	
Vertical jump test post-values	Group A (Tic-tac-toe exercise)	21	18.7905 $\pm$ 2.88546	.005
	Group B (squat jumps)	21	20.8810 $\pm$ 1.39557	

Data analysis of both groups showed an increase in mean value from pre-test to post-test for the vertical jump test. This showed the effectiveness of both regimens, but the greater increase in mean value for squat jumps showed that it is more effective than tic-tac-toe drills. This table showed that significant difference between the groups for 30-meter sprint test as the sig.(2-tailed) value was 0.005, which was less than the 0.05 set value.

CHAPTER # 6
DISCUSSION

Discussion

A randomized clinical trial was conducted with the purpose of seeing the comparative effect of tic-tac-toe drills versus squat jumps on speed, agility, and explosive power among fast bowlers. Two groups, group A and group B, followed the tic-tac-toe drills and squat jumps with whole body dynamic stretching as a baseline. The results of the study showed that both techniques were statistically significant for improving speed, agility, and explosive power, also, squat jumps showed greater improvements. Both the tic-tac-toe and squat jumps were equally effective in improving the agility T test, while squat jumps regimen was superior to tic-tac-toe exercise in improving the 30-meter sprint test and vertical jump test.

Sitanggang D.L.F. conducted a study in 2024 on the tic-tac-toe exercise model on basketball players to see its effect on speed, agility, and accuracy. This research adopted the model based on the Analysis, Design, Development, Implementation, Evaluation method for the development of this regimen. This study used valid and reliable tests involving the 30-meter sprint test, Illinois test, and accuracy test norms to interpret the results of speed, agility, and accuracy. The analysis of this study showed that tic-tac-toe exercise effectively increases speed, agility, and accuracy. The p-value for the Illinois test, the 30-meter sprint test were 0.002 and 0.001, respectively. The p-value of agility T-test and 30-meter sprint test after following tic-tac-toe exercise in our study was <0.001 , which means that the above study supported our study.

The effectiveness of squat jumps in improving speed, agility, and explosive strength performance of athletes was the subject of a study by H. Haung and Y. Haung in 2023. An 8-week plyometric training involving 15 male elite players received 24 plyometric training courses that were given three times per week for an 8-week program. All the given regimen courses were executed by pre- and post-testing. The 20-meter sprint test and T agility test showed a decrease in the mean value in post-testing, with a significant value of 0.001, which was less than 0.05. The p-value for these variables in our study was also less than 0.001. This study supported our training regimen that squat jumps improved the physical performance of athletes by targeting the same variable we included in our research design.

Literature supported the squat jumps effectiveness on physical training components i.e., speed, agility, and explosive power. Isaac and his colleagues in 2022 through their study endorsed the effect of squat jumps. Through their study, they showed that squat jumps significantly improved the lower limb explosive power, through which athletes improved a lot in sports by gaining rapid acceleration and vertical jumping. As the p-value was significant for the conducted tests for speed, agility, and explosive power, which supported our study, as the significant value for our study was also <0.001 .

Similarly, in support of our research, an 8-week plyometric training involving the pre- and post-testing design by S. Kosova and Co, showed the effect of squat jumps on jump height, speed, and explosive power. This study was carried out with a sample size of 26 fencers who followed an exercise regimen 3 times per week for an 8-week training program. We included the fast bowlers in our study to see the effect of squat jumps with the exercise model involving 3 times per week training for a 6-week program. The study stands out in favor of our training regimen, with a p-value less than 0.05 for conducted tests that squat jumps positively contributed to speed, agility, and explosive power.

Eli Arianda did the study to see the effect of plyometric training on increasing the speed and explosive power among the chosen sample. 60 students were pre-tested using 30-meter sprint test and standing broad jump. After the training, the tests showed a significant improvement in both speed and explosive power, with the p-value of 0.002 and 0.034, which were less than 0.05. The results were in favor of our study, as the p-value for the speed tests and explosive power testing methods in our study was also less than 0.05. Thus, this study supported our research.

In 2024, a study was conducted to see the effectiveness of squat jumps and countermovement jumps in improving vertical jump performance. 74 athletes participated in this study and were pre- and post-tested using the vertical jump test. The two treatment groups followed their regimens, and the results showed a significant value of less than 0.05, showing there was an improvement in vertical jump ability in both groups. This study supported our study as the results were the same in our study, that the p-value was less than 0.05 for the vertical jump test after following the squat jump regimen.

6.1 Conclusion

The results of the study showed that for improving the 30-meter sprint test and vertical jump test, the squat jumps regimen was superior to tic-tac-toe exercise, while for the agility T-test, the squat jumps regimen and tic-tac-toe exercise showed equal effects.

6.2 Limitations

- This study was limited to Pakistan cricket academies and clubs.
- The population of this study involved only male fast bowlers aged 18-30 years.

6.3 Recommendations

- It is recommended that fast bowlers must include the exercise in their training program for improvement in athletic performance.
- Researchers should conduct this research on other athletes, such as volleyball players, basketball players, and football players.
- Researchers should conduct this research on a higher level with a greater sample size to see its long-lasting effects.

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CONSENT FORM

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course the study without consequence. I understand that any information resulting from this study will be strictly confidential. I realized that I may ask for further information about this study if I wish to do so at any time.

I have received a copy of this consent form for my own records. I agree to participate in this study.

Subject Name _____

Age _____

Date _____

Telephone _____

Signature _____

Profession _____

QUESTIONNAIRE

Demographic Data:

Name: Age:

Gender: ☐ Male ☐ Female

Playing time one year: ☐ Yes ☐ No

Occupation:

Phone number:

History of any systemic illness: ☐ Yes ☐ No

History of fracture/traumatic injury (in the last 6 months): ☐ Yes ☐ No

Any neural injury (in the last 3 months): ☐ Yes ☐ No

History of any sprain or strain (in the last 3 months): ☐ Yes ☐ No

	Tic-Tac-toe exercise		Squat jumps	
Measurement Tools	Pre-test value	Post-test value	Pre-test value	Post-test value
T-test				
30-meter run test				
Vertical Jump test				



Fast bowlers



Tic-tac-toe frame



Warm-up



Performing tic-tac-toe exercise